

ThermoTumbler: A Tumbler-type Device that Changes Flavor Perception by Controlling Temperature to the Lower Lip

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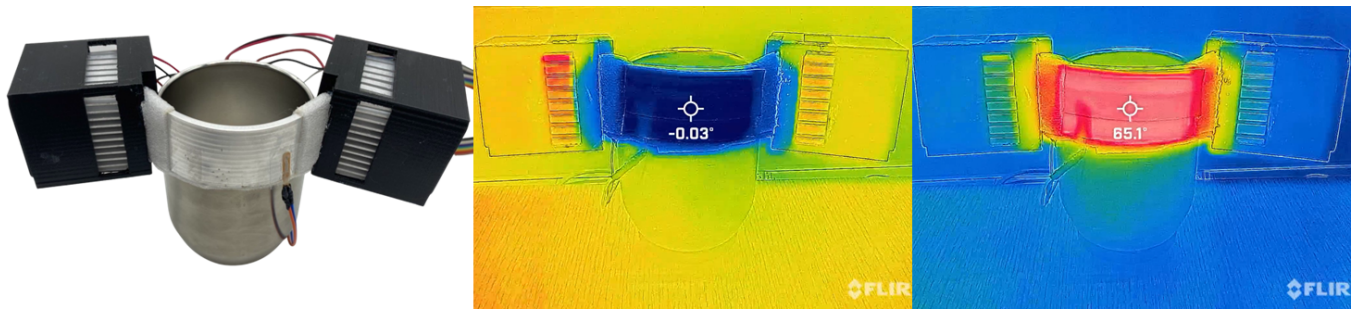


Figure 1: ThermoTumbler (left: overview, middle: when the cooling drinking spout, right: when the warmed drinking spout)

ABSTRACT

We propose *ThermoTumbler* that realizes affects the user's flavor perception, including "in-mouth aroma," "aftertaste," "intensity of taste," "throat feeling," "deliciousness," "comfortable," "sweetness," "saltiness," "sourness," "bitterness," and "umami," by presenting temperature around the lower lip without altering the beverage. The ThermoTumbler utilizes Peltier devices and heat pipes to precisely regulate the temperature at the drinking spout within the range of 0 to 65 °C, ensuring that it presents a specific temperature to the user's lower lip. We investigated the changes in flavor perception resulting from temperature to the lower lip using the ThermoTumbler. The evaluation experiment's results showed that participants' perceptions of the beverage's "temperature," "throat feeling," "deliciousness," and "comfortable" varied according to the temperature cues presented at the drinking spout. In addition, it was shown that temperature stimulation on the lower lip can accentuate a beverage's five basic tastes.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction paradigms*; Mixed / augmented reality ; • **Computing methodologies** → **Computer graphics**; *Graphics systems and interfaces*; Virtual reality.

KEYWORDS

Flavor perception; Human Food Interaction; Pseudo-physiological reaction; Thermal sensation

ACM Reference Format:

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1 INTRODUCTION

Meals are delightful experiences that bring the greatest joy and happiness, and savoring them contributes to a fulfilling life[17, 29]. On the other hand, improving dietary habits is crucial for preventing lifestyle-related diseases and obesity, and controlling excessive salt intake is linked to preventing hypertension. Therefore, managing one's diet is indispensable for a healthy lifestyle. However, it often requires moderate salt and sugar intake and exercising restraint with favorite foods and beverages. Consequently, research is advancing in presenting artificial tastes (five basic tastes) to support a healthy lifestyle.

Some methods combine chemical substances to reproduce taste artificially, but the difficulty of handling these chemicals makes them impractical for everyday use. In recent years, research has been conducted on various taste enhancement techniques that do not rely on chemical substances. For example, studies have used electrical stimulation on the tongue to present five basic tastes as a form of taste perception. While taste presentation is possible with electric stimulation on the tongue, altering non-taste aspects of flavor perception, such as "in-mouth aroma," "aftertaste," "intensity of taste," "throat feeling," "deliciousness," and "comfortable," is problematic.

Recently, the cross-modal technology that emphasizes sensory perception by combining multiple senses, such as visual and auditory stimuli, has gained attention and has been used in various research fields. Some researchers clarified the effects of temperature on flavor perception, including presenting temperature to the tongue and nose[12, 38, 42]. However, they have yet to be conducted on individually controlling the temperature of areas around the mouth, such as the lips and the temperature of the food or drink. For example, during chewing, the lower lip comes into contact with cutlery and tableware, suggesting that the temperature sensation of the lower lip is also important for flavor perception. Therefore, we focused on drinking spout temperature on flavor perception. This study focuses primarily on beverages because of the lack of chewing effects and the ease of standardizing parameters such as "throat feeling" without interference.

To investigate the effect of lower lip temperature on flavor perception, we have developed "ThermoTumbler" (Figure 1), which is capable of changing the drinking spout a specific temperature to affect the user's flavor perception. We will then use the ThermoTumbler to verify the effects of these temperature stimuli on flavor perception.

Here, we define "in-mouth aroma," "aftertaste," "intensity of taste," "throat feeling," "deliciousness," "comfortable," "sweetness," "saltiness," "sourness," "bitterness," and "umami" as the "flavor perception" perceived by users.

2 RELATED WORK

Compounds are often used as a method to reproduce taste sensations[3, 18, 21]. However, studies involving chemical compounds present challenges related to the management and optimal formulation of these compounds, making their implementation difficult. In addition, the long-term use of artificial chemicals such as monosodium glutamate may have potential adverse effects on the human body [22].

Therefore, some researchers proposed methods using electrical stimulation is applied to the tongue for present the basic taste perception [1, 20, 23, 32]. In taste perception, however, while electrical stimulation control is more straightforward than chemical control.

Other researchers have proposed various methods to modify flavor perception by employing sensory interactions, including visual, auditory, olfactory, and gustatory stimuli[15, 24–28, 31, 39–41, 45]. In addition, they has been shown that tactile sensations around the lower lip or on the hands (affected by the shape or type of cup) can also alter flavor perception[2, 4–6, 10, 19, 30, 33, 35, 36, 43].

Several studies have demonstrated that food temperature can affect flavor perception[8, 14, 16, 34, 37, 44]. In other words, by controlling the temperature of the food itself, that is, by presenting a certain temperature to the oral cavity through the food, it is possible to change flavor perception.

Applying such characteristics, some methods of flavor perception use temperature stimuli around the nose and tongue and food temperature[7, 38, 42]. However, research has yet to be conducted on individually controlling the temperature of areas around the mouth, such as the lips, and the temperature of food or drink. For example, during chewing, the lower lip comes into contact with cutlery and tableware, suggesting that the temperature sensation

of the lower lip is also important for flavor perception. Therefore, we focused on the lower lip temperature on flavor perception.

Based on previous research suggesting changes in flavor perception due to temperature stimuli on the tongue and around the nose [7, 12, 38], this study aimed to investigate the effects of temperature stimuli around the lower lip. "Affecting Tumbler"[38] has been shown to affect comfort and aftertaste by presenting temperature to the nose area without changing the flavor perception. Based on this finding, we focus specifically on drinking. It proposes the "ThermoTumbler," a device that provides optional temperature stimuli to the lower lip simultaneously with the consumption of a beverage, to explore the aspects above. In our experiment, we aimed to evaluate changes in flavor perception due to temperature differences in the drinking spout's temperature, which was different from the beverage's temperature while consuming beverages under three temperature conditions (cooled, room temperature, warmed).

3 DESIGN AND IMPLEMENTATION

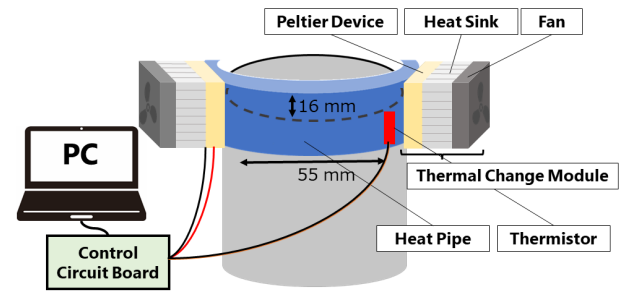


Figure 2: Composition of ThermoTumbler

Figure 2 shows the configuration of ThermoTumbler. ThermoTumbler consists of a tumbler, an aluminum heat pipe, two thermal change modules, a thermistor to obtain the temperature of the drinking spout, a control board, and a PC. The aluminum heat pipe is attached to the drinking spout, which efficiently transfers heat. Two thermal change modules at both ends of the heat pipe vary the temperature presentation to the lower lip. The thermal change module comprises a Peltier device, a heat sink, and a fan to warm or cool by flowing current in the forward or reverse direction from the motor driver to the Peltier device. We arranged a heat sink and a fan to cool the Peltier device, which releases the heat amount of absorbed heat and consumed power to the heat radiation side.

We implemented a prototype system of ThermoTumbler based on the configuration shown in Figure 2. The weight of the ThermoTumbler we created was 400 grams. We adopt a vacuum insulation structure in the tumbler to keep the beverage's temperature and prevent the heat from being transmitted to the outside. For the drinking spout, we chose an aluminum heat pipe with high specific heat and thermal conductivity from materials used in tableware and cutlery.

We inserted insulation material between the tumbler and the heat pipe to prevent the temperature of the drinking spout from being transmitted to the tumbler. We also used a neoprene-based adhesive to prevent the beverage from seeping through the gap. We implemented an Arduino and a motor driver on the control board

for temperature control. Peltier devices are mounted to control the temperature of the drinking spout, and a thermistor provides feedback to the Peltier devices. In addition, a PID control system is adopted for driving the Peltier devices to change the temperature of the drinking spout.

After implementation, we conducted experiments to evaluate ThermoTumbler's performance for cooling and warming capabilities. The temperature of the drinking spout can be precisely varied within the range of 0 °C to 65 °C by continuously warming and cooling it for 5 minutes.

4 USER EXPERIENCE

This study aims to assess changes in flavor perception due to temperature differences by placing beverages into the mouth at three temperature conditions (cool, room temperature, and warm), where the temperature of the beverage differs from the temperature of the drinking spout.

4.1 Experimental Design

Previous research has shown that changes in flavor perception due to food temperature vary depending on food type[8, 14, 16, 34, 37, 44]. Furthermore, the "Affecting Tumbler" study examined the effects on flavor perception of temperature stimuli applied to the skin around the nose using orange juice and apple juice[38]. Inspired by these studies, in this experiment, we used orange juice, which provides "sourness," and apple juice, which provides "sweetness" (both produced by Ehime beverage Co., Ltd.), along with green tea, which provides "bitterness" (produced by HARUNA Produce Co., Ltd.), and water, which is tasteless and odorless.

The combination conditions of drinking spouts and beverage temperatures (nine conditions) are shown in Table 1. In the previous "Affecting Tumbler" study, the effect on flavor perception was investigated by varying the skin temperature around the nose by ± 2 °C from room temperature[38]. Taking this as a reference, in the current experiment, three temperature conditions we set for the drinking spout: the same temperature as the beverage, -2 °C for the cooling condition, and +2 °C for the warming condition. The temperature of the beverage was three conditions: cooling, room temperature, and warming. The cooling beverage was set at 5 °C, and the room temperature beverage was set to 24 °C, aligning with the ambient temperature. For the warming condition of beverages, particularly considering the potential risk of causing significant tissue damage with temperatures exceeding 50 °C on the lower lip [9], the maximum temperature for the lower lip was set at 47 °C. Therefore, we set the warmed beverages at 45 °C. The aroma and basic taste of food and beverages can change with temperature variations [8, 14, 16, 34, 37, 44]. Although it is uncommon to warm juices, this experiment included a condition for warming juice to explore its potential effects. We regulated the temperature of the cooled beverages using a refrigerator. We managed the temperature of the warmed beverages by first warming them in a microwave oven and then using a cup warmer to keep the temperature.

We conducted the evaluation using the method of magnitude estimation using a scale with one as the standard on a drinking spout at the same temperature as the beverage. We asked participants to

compare the intensity of the following factors in subsequent tastings. The evaluation items are as follows: "beverage temperature," "in-mouth aroma," "aftertaste," "intensity of taste," "throat feeling," "deliciousness," "comfortable," "sweetness," "saltiness," "sourness," "bitterness," and "umami." The temperature evaluation included separate assessments for "warm" and "cool."

We included the "beverage temperature" category to assess whether there were changes in beverage temperature perception due to temperature cues applied to the lower lip. We included the categories "in-mouth aroma," "aftertaste," and "intensity of taste" to obtain objective ratings of texture. In addition, we included the categories of "deliciousness" and "comfortable" to assess participants' personal preferences. In addition to these six texture-related items, we created five items to assess basic taste. In addition, we asked for participants' impressions of each comparison stimulus through open-ended responses.

4.2 Experimental Procedure

The participants in the experiment consisted of 18 individuals (9 males, 9 females) with an average age of 22.1 ± 1.3 years old. Specifically, for the green tea condition, there were 20 participants (10 males, 10 females) with an average age of 22.5 ± 1.1 years old. We explained the experiment to the participants before they began and obtained their signatures on an informed consent form. We informed the participants in advance that the experiment involved tasting and rating beverages and that they were free to withdraw from the study at any time. During this explanation, we confirmed that participants had no aversion to the beverages provided and were free of allergies. We asked participants to subjectively rate their current level of throat dryness using the Visual Analog Scale method on a 100 mm line to minimize the potential impact of throat dryness on flavor perception ratings before the start of the experiment. The scale ranged from "not dry" at 0 to "very dry" at 100. If the rating was 70 or higher, the participants were given a small amount of water before the experiment.

First, we presented the participants with a tasting using a drinking spout at the same temperature as the beverage (standard). Subsequently, participants orally rated the comparison stimulus relative to the standard. We conducted this as one set and performed six sets (one for each of the six conditions). We measured the drink's temperature immediately before presentation using an infrared radiation thermometer; each cup contained 10 g of the drink. To account for order effects, we presented combinations of drinking spouts and beverage temperatures in randomized conditions. We did not inform the participants about the combination of drinking spouts and beverage temperature. We informed participants that they could taste the beverages as many times as they wished, that they were not obligated to consume all of the liquid in the cup, and encouraged them to bring the cup to their mouths as soon as they were handed to them. In addition, to eliminate lingering tastes, we instructed the participants to take a sip of water before tasting a different condition, especially if the beverage was something other than water. We instructed the participants to drink by placing the heat pipe attached to the cup against their lower lip to achieve temperature stimulation of the lower lip. The ethics review board of our university approved this study (approval number: H22-032).

Table 1: Combination of beverage and drinking spout temperature

	cooling drinking spout	standard	warmed drinking spout
cooling beverage (5 °C)	3 °C	5 °C	7 °C
room temperature beverage (24 °C)	22 °C	24 °C	26 °C
warmed beverage (45 °C)	43 °C	45 °C	47 °C

The room temperature during the experiment was an average of 24.5 ± 0.5 °C, and the humidity was 11 ± 11 %.

4.3 Results and Discussion

We investigated whether the temperature difference between drinking spouts and beverages affected flavor perception. We standardized the data for each participant with a mean of 0 and a standard deviation of 1 because the scale varied among individuals. As a standardization method, we calculated the mean and standard deviation of all the data for each participant. We then subtracted the mean and divided it by the standard deviation of each score. Shapiro-Wilk tests on the standardized rating data indicated that none of the rating items followed a normal distribution. Therefore, we conducted the nonparametric Friedman test and multiple comparisons using the Bonferroni method to investigate the significance of the differences with each rating dataset. The differences between the medians of the standardized rating data for each beverage and the standard median are shown in Figures 3.

For some factors, there was no significant difference in the drinking spout temperature when the difference was ± 2 °C. However, in cases where the drinking spout temperature was ± 4 °C (cooling drinking spout and warmed drinking spout), it was evident that this affected flavor perception. This suggests that changing the temperature range, such as cooling or warming, by more than 4 °C compared to the room temperature drinking spout can lead to further changes in flavor perception.

The water exhibited fewer changes in flavor perception than juice and green tea due to the drinking spout temperature. For this reason, water has fewer components and is nearly tasteless and odorless compared to juice and tea. On the other hand, each juice and green tea has different components, leading to variations in flavor perception induced by drinking spout temperature unique to each beverage. In the following sections, we will discuss each flavor perception evaluation separately.

4.3.1 The results and discussion of "beverage temperature". The warmed drinking spout was perceived as significantly warmer than the standard in the warmed juice condition. In contrast, there was no significant difference in beverage temperature perception in the warmed water and green tea conditions. For this reason, the effects on temperature perception when the temperature of the drinking spout changes are likely to depend on the beverage ingredients.

In all cooling beverage conditions, the cooling drinking spout perceived the beverages as significantly colder than the standard. In the case of the cooling drinking spout, the sensation of something colder than the beverage presented to the lower lip may lead to the perception that the beverage itself is colder than the standard. On the other hand, in all colling beverage conditions, the warmed drinking spout perceived the beverages as significantly colder than

the standard. Because the lower lip to be initially warmed if the drinking spout presents a temperature warmer than the beverage, the cooling beverage could be felt colder than the standard.

4.3.2 The results and discussion "in-mouth aroma". In the warmed green tea condition, it was evident that the warmed drinking spout tended to increase "in-mouth aroma" compared to the standard. The reason is that warming green tea made the aroma more pronounced. In addition, warming the drinking spout further enhanced the "in-mouth aroma."

4.3.3 The results and discussion "aftertaste" and "intensity of taste". In the orange juice and apple juice conditions, there was a significant difference in "aftertaste" based on the temperature of the drinking spout. However, a significant difference in the "intensity of taste" rating was observed only in the case of orange juice. From these results, the changes in "aftertaste" and "intensity of taste" due to drinking spout temperature may vary depending on the beverage's inherent flavor perception. On the other hand, the combination of room-temperature orange juice with a cooling drinking spout showed a significantly increased "intensity of taste" compared to the combination with a warmed drinking spout. The combination of warmed green tea with a warmed drinking spout showed a significantly increased "intensity of taste" compared to the combination with a cooling drinking spout. While there was no significant difference at a temperature difference of 2°C (cooling drinking spout vs. standard, warmed drinking spout vs. standard), there was a significant difference at a temperature of 4°C. From these results, changing the setting temperature of the spout can indeed change the perception of "intensity of taste."

4.3.4 The results and discussion "throat feeling," "deliciousness," and "comfortable". Cooling drinking spout for cooled and room temperature juice and green tea conditions showed a tendency to increase "throat feeling," "deliciousness," and "comfortable." By altering the "throat feeling" through temperature presentation drinking spout, it was possible to improve the perceived "deliciousness" of the beverage. In addition, the improved perception of the beverage's flavor perception likely contributed to an increased sense of "comfortable."

4.3.5 The results and discussion five basic tastes. The drinking spout temperature changed "sweetness" for apple juice and "sourness" for orange juice. Because apple juice tends to be sweeter than orange juice and orange juice tends to be more acidic than apple juice [11], the drinking spout temperature changes in "sweetness" for apple juice and "sourness" for orange juice. In the case of orange juice, there was a tendency for increased "sweetness" when the drinking spout was cooled. Fruit juice quality varies based on the sugar-acid ratio [13]. Since orange juice is known for its distinct "sweetness" and "sourness," cooling the spout under orange

		beverage temperature (warm)	beverage temperature (cool)	in-mouth aroma	aftertaste	intensity of taste	throat feeling	deliciousness	comfortable	sweetness	saltiness	sourness	bitterness	umami
Cooled (5°C)	Orange Juice		Wds-std* Cds-std** Wds-Cds**		Cds-std†		Cds-std* Wds-Cds*	Cds-std**	Cds-std* Wds-Cds†			Wds-std*		
	Apple Juice		Wds-std** Cds-std**				Cds-std* Wds-std†	Cds-std†	Cds-std† Wds-Cds†	Cds-std†				
	Water		Wds-std** Cds-std**				Cds-std†							
	Green Tea		Wds-std* Cds-std**				Cds-std** Wds-std†	Cds-std* Wds-Cds†					Cds-std* Wds-Cds†	
Room temperature (24°C)	Orange Juice		Wds-std† Cds-std*			Wds-Cds*	Cds-std† Wds-Cds*	Cds-std† Wds-Cds*	Cds-std* Wds-Cds*	Cds-std†		Wds-std†		
	Apple Juice		Cds-std*		Cds-std*		Cds-std†	Cds-std**		Cds-std*				
	Water		Wds-std* Cds-std*											
	Green Tea						Cds-std†	Cds-std†	Cds-std*					
Warmed (45°C)	Orange Juice	Wds-std*			Cds-std* Wds-Cds†	Cds-std*								
	Apple Juice	Wds-std*												
	Water	Wds-std†												
	Green Tea			Wds-std†		Wds-Cds*		Wds-std*	Wds-std†				Wds-std†	

**: $p < 0.01$, *: $p < 0.05$, †: $p < 0.1$

Wds:warmed drinking spout(beverage+2°C), Cds:Cooling drinking spout(beverage-2°C), std:standard(the same temperature as the beverage)

Figure 3: Results of flavor perception

juice conditions likely influenced changes in both "sweetness" and "sourness."

In the cooled green tea condition, the cooling drinking spout significantly decreased perceived "bitterness" compared to the standard. In the warmed green tea condition, there was a tendency for the warmed drinking spout to increase perceived "bitterness" compared to the standard. Green tea is a beverage known for its tendency to evoke a sense of "bitterness." Therefore, we considered that the drinking spout temperature influenced the perceived "bitterness" alteration.

5 LIMITATION AND FUTURE WORK

As a result, we found that the presentation of temperature to the lower lip by the ThermoTumbler does indeed affect flavor perception. However, the mechanisms behind these phenomena remain unclear. Previous research has shown that flavor perception can be influenced by the temperature of foods [8, 14, 16, 34, 37, 44]. We consider that flavor perception changes with the proposed method depending on the beverage's temperature. Therefore, it is necessary to study the influence of temperature on the components of the beverage.

In addition, individual preferences and cultural factors may also influence flavor perception changes with the proposed method. In this experiment, some participants were not accustomed to consuming warmed beverages, found 45 °C warmed beverages lukewarm, or did not prefer 5 °C cooling beverages. Therefore, we plan to increase the number of participants and conduct experiments to elucidate the relationship between individual differences and cultural variations in flavor perception.

The results indicate that the sensations of "throat feeling," "deliciousness," and "comfortable" are improved by the temperature applied to the lower lip, regardless of the temperature of the beverage. From these results in disaster scenarios, where there may not be the flexibility to cool or warm beverages, the method significantly impacts flavor perception. In addition, temperatures around 24 °C are conducive to rapid changes, allowing us to complete temperature changes just before using the tumbler. That allows for seamless changes in flavor perception without causing discomfort. On the other hand, the experimental results show that when the temperature applied to the lower lip is 5 °C±2 °C, the "throat feeling" is improved compared to the 24 °C±2 °C condition. From these results, this temperature range is desirable when paired with beverages that benefit from a colder texture, such as juices or beer. Unlike traditional establishments that rely on freezing glasses for extended periods, the ThermoTumbler offers the flexibility to enhance smoothness through chilling and manipulate the temperature, thereby altering flavor perception in various ways.

6 CONCLUSION

In this study, we aimed to change flavor perception without changing the beverage by providing temperature stimulation to the lower lip. We proposed the use of the ThermoTumbler for this purpose. Using the ThermoTumbler, we evaluated whether flavor perception could be altered by providing temperature stimulation to the lower lip while consuming a beverage.

We used orange juice, apple juice, water, and green tea in the experiment. We presented three conditions for each beverage: cooling, room temperature, and warming, and the drinking spout conditions:

the same temperature as the beverage and a temperature within ± 2 °C.

The experiment results showed that temperature stimulation on the lower lip can accentuate a beverage's five basic tastes. In addition, it was evident that the proposed method could influence the perception of beverage temperature and affect flavor perception. The effect of lower lip temperature stimulation on flavor perception varied depending on the type of beverage.

In this experiment, we presented a constant temperature; however, we plan to verify the effect on flavor perception by changing the temperature during drinking. In addition, we plan to increase the number of participants and the variety of beverages to investigate further the relationship between lower lip temperature presentation and flavor perception. Furthermore, we aim to implement temperature-controlled utensils, such as spoons, and explore the application of the proposed method to food other than beverages. By using these devices, we expect smoother texture, enhanced flavor, and a more pleasant eating experience, allowing individuals to enjoy even foods they may find unpalatable. In addition, if it becomes possible to create the illusion of preferred temperatures for room-temperature foods, this could prove valuable in situations such as disasters with limited cooking capacity.

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